

SCALABILITY ROADMAP & FUTURE SCOPE

Developer: [Left Blank] | OptiCrop Project Documentation

Repository: <https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine>

1. Planned System Enhancements

- IoT Integration: Connect soil sensors to collect NPK, pH, and humidity data automatically.
- Deep Learning Disease Detection: Build a PyTorch CNN to identify crop leaf anomalies from images.
- Weather API Integration: Integrate weather forecast services to schedule irrigation automatically.
- SaaS Multi-tenancy: Add authentication, subscriptions, and user management for agribusinesses.

Opticrop - Scalability & Roadmap

- Short-term (3 months)
 - REST API for IoT soil sensors (real-time NPK feeds)
 - REST API for weather services (real-time NPK feeds)
 - Multi-language support (Hindi, Telugu, Tamil)
- Mid-term (6 months)
 - Weather API integration for automated irrigation alerts
 - Mobile-responsive PWA with offline mode
- Long-term (12 months)
 - Weather API integration for automated irrigation alerts
 - Mobile-responsive PWA with offline mode
 - Leaf-disease CNN (PyTorch) via camera upload
 - SaaS multi-tenant login (farmer history & trend analytics)
- Long-term (12 months)
 - Govt. API linkage for subsidy eligibility checks
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